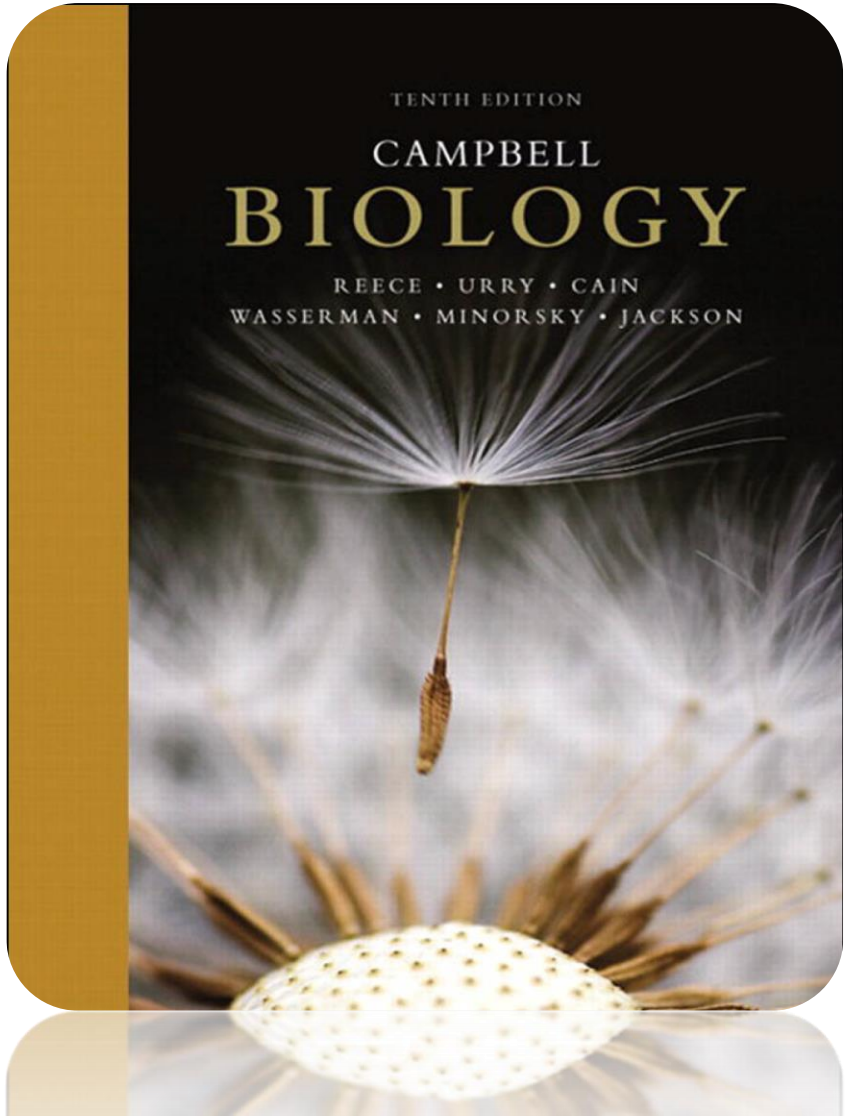
LECTURE-1

General outline of the course

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Reference Book



Bio-101: Foundations of Biology (FOB)
Compulsory course for CSBS.

Prescribed Book: Campbell's Biology, Tenth Ed.
(or) Global Edition

Available in the library:
https://library.iiitd.edu.in/cgi-bin/koha/opac-detail.pl?biblionumber=25258

Slides: All lecture slides and other materials will be uploaded in the course webpage.

Why Foundations of Biology?

CSBS

- Biology
  - Cell-biology & Biochemistry
  - Genetics & Molecular biology
  - Ecology & Evolution

  ❖ Introducing to broad basics of biological sciences.

  ❖ The course will introduce to you a spectrum of topics.

  ❖ Pre-requisite to doing well in other subsequent courses in the degree program.

- Mathematics
  - Probability & Statistics
  - Calculus

  ❖ Refreshing basic-101 course you have done so far.

  ❖ Refreshing school calculus is sufficient to solve simple problems like data analysis, visualisation, and building simple models in this course.

- Algorithms / programming

  ❖ Python / R for visualisation

  ❖ Mostly used like a calculator and for coding small things.

  ❖ Important for assignments

Objectives and Course outcome

❖ The objectives of this course are

1. To introduce the basics of biology regarding cellular mechanisms, Genetics, and Evolution.
2. Wide spectrum of topics will be covered in this course. This includes basic biochemistry, cell biology, genetics, molecular biology, ecology, and evolution.
3. It has both quantitative and qualitative aspects.

❖ The broad course outcomes are

1. to relate structure to function.
2. to cover the basics of Genetics.
3. to cover the basics of molecular evolution.

Broad topics covered in this course

Unit 1: The Chemistry of Life *The Chemical Context of Life, Water and the Fitness of the Environment, Carbon and the Molecular Diversity of Life, The Structure and Function of Macromolecules*
Unit 2: The Cell *A Tour of the Cell, Membrane Structure and Function, An Introduction to Metabolism, Cellular Respiration: Harvesting Chemical Energy, Photosynthesis, Cell Communication, The Cell Cycle*
Unit 3: Genetics *Meiosis and Sexual Life Cycles, Mendel and the Gene Idea, The Chromosomal Basis of Inheritance, The Molecular Basis of Inheritance, From Gene to Protein, The Genetics of Viruses and Bacteria, Eukaryotic Genomes: Organization, Regulation, and Evolution, DNA Technology and Genomics, The Genetic Basis of Development*
Unit 4: Mechanisms of Evolution *Descent with Modification: A Darwinian View of Life, The Evolution of Populations, The Origin of Species, Phylogeny and Systematics*
Unit 5: The Evolutionary History of Biological Diversity *The Tree of Life: An Introduction to Biological Diversity, Prokaryotes, Protists, Plant Diversity I: How Plants Colonized Land, Plant Diversity II: The Evolution of Seed Plants, Fungi, An Introduction to Animal Diversity, Invertebrates, Vertebrates*

Evaluation

Test No	Type of test	Duration	Types of questions	Weightage
1	Assignments-I/II (Prior to Midsem)	2-3 days	Qualitative/ Quantitative	20
2	Assignments (Prior to endsem)	2-3 days	Quantitative / Qualitative	20 (Submission: 1 prior to mid-sem / 1 after the mid-sem)
3	Midsem	1 hr	-do-	20
5	Endsem	2 hr	-do-	40
6	Tutorials	-	Quantitative	None. Tutorials will help to reinforce the concepts learnt in the class

Lectures: M/W/Thu, C-02 (Old academic)(11 - 12 AM)
Tutorials Friday, C-02 (R & D block) (8:30 - 9:30 AM).

Key concepts

KEY CONCEPTS

1.1 The study of life reveals common themes

1.2 The Core Theme: Evolution accounts for the unity and diversity of life

1.3 In studying nature, scientists make observations and form and test hypotheses

1.4 Science benefits from a cooperative approach and diverse viewpoints

Unifying themes about life

- What is life and how is it related to evolution?

- Nothing in Biology Makes Sense Except in the Light of Evolution" is a 1973 essay by the evolutionary biologist Theodosius Dobzhansky

- What is life? A philosophical question. We can say dogs and cats are alive, but not computers / I-phones. However, it can be summarized by understanding broadly four major themes

- Four major themes about studying life comes out.

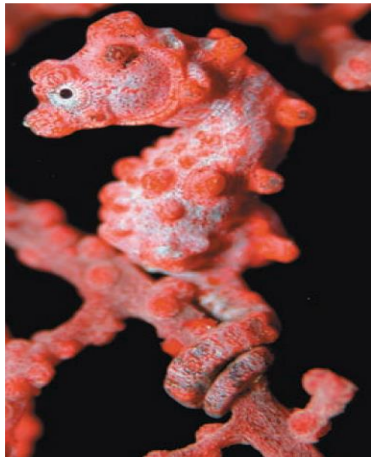
1. Theme-1: Organization
2. Theme-2: Information
3. Theme-3: Energy and Matter
4. Theme-4: Interactions

All these themes plays a role in shaping the evolution and adaptation of under various conditions.

Properties of life

❖ Rather than asking what life is, we will look into the characteristic features of life. They are the following.

Order. Highly ordered structure



Evolutionary adaptation. sea horse camouflages to its environment for survival.

Reproduction. Organisms (living things) reproduce their own kind.

Regulation of blood flow in jackrabbit's ears to maintain a constant body temperature by adjusting heat exchange with the surrounding air.

Response to the environment. Venus flytrap traps the fly rapidly in response to the environmental stimulus

Growth and development. Genotype-phenotype relationship. Controls pattern.

Energy processing. The nectar obtained from flowers will use chemical energy stored in its food to power flight and other work.

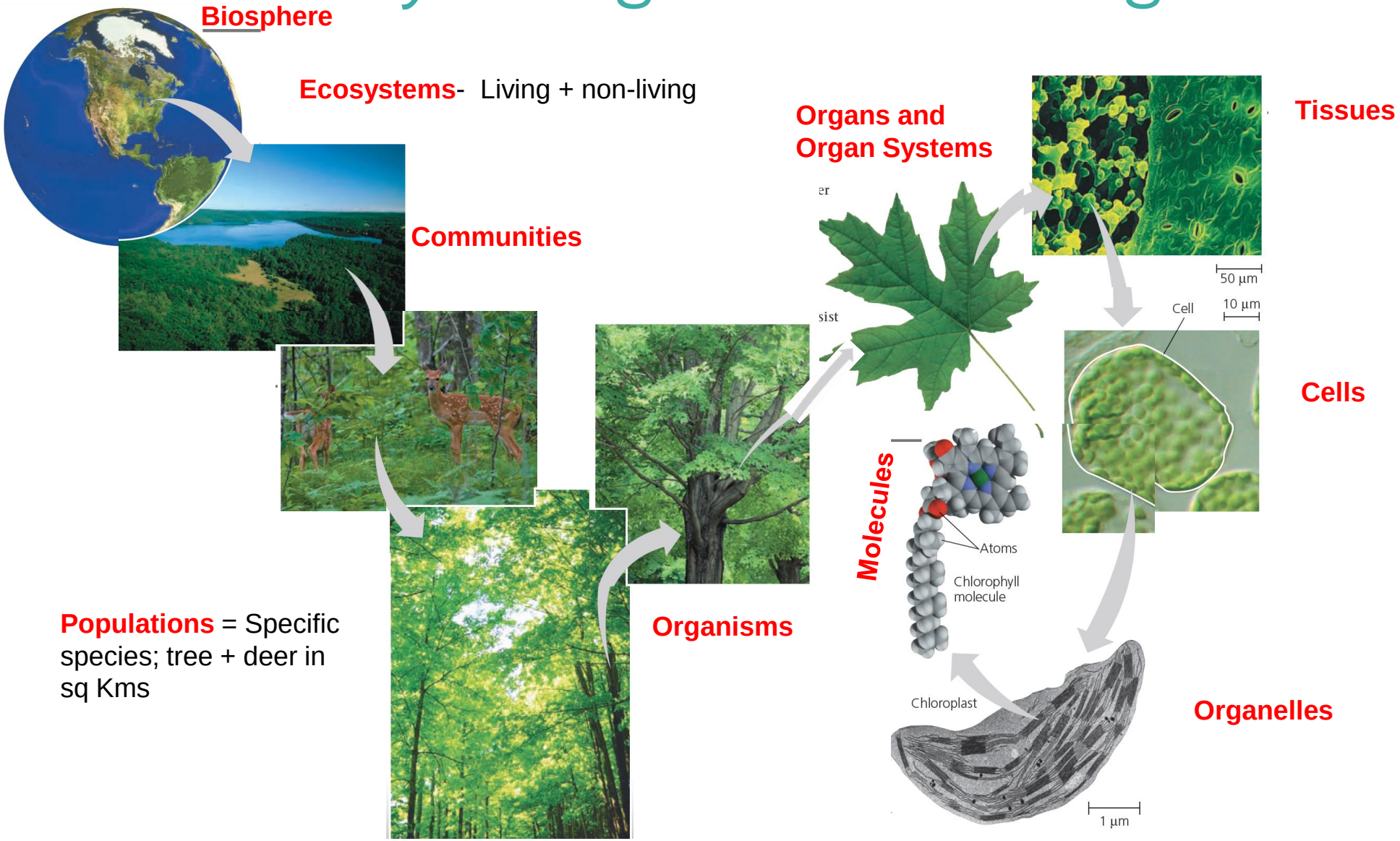
Complex and complicated systems

Difference

Engineering systems	Biological systems
Complicated i.e., do not have ability to evolve and learn	Complex i.e., have ability to evolve and learn
Closed systems. No influx and outflux	Open systems, far from equilibrium
Cannot adapt to perturbations on their own	Can adapt to various perturbations and can gauge the environment
No emergence, because it cannot evolve and learn	Can display self-organization without a central organizing authority i.e., emergence
The role of small pieces in the system can be understood and the whole system can be reassembled from its parts	The role of a small piece in the whole system cannot be determined in isolation

Ref: J. M. Ottino (2004) Engineering complex systems, Nature, 409, 399.

Macroscopic to microscopic scales of life: Hierarchy of organization of organisms



- Reverse the arrows (zoom out) and move from molecules to the biosphere.
- Novel properties arise at each higher level, which was not present at the preceding level.
- Such emergent properties result from the specific arrangement and interactions of component parts.
- In emergent property, the whole of which is more than the **sum** of its parts.

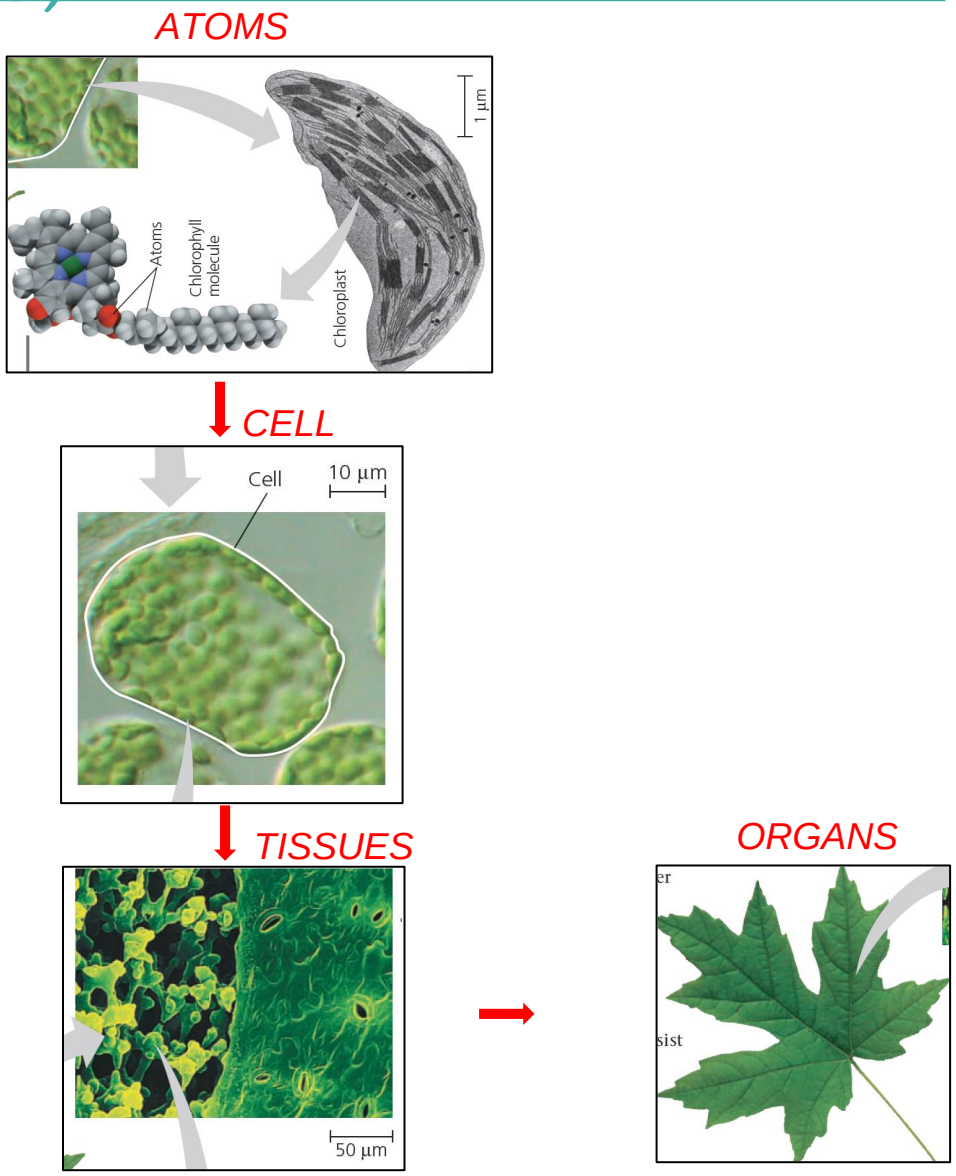
Hierarchy in the organisation of an organism (Vertical Scale)

- The chemical level, the most basic level of organisation, includes atoms and molecules.

- At the cell level, many atoms and molecules associate to form *cells*. However, the cell's emergent properties make it the basic structural and functional unit of life, the simplest component of living matter that can carry on all the activities necessary for life.

- During the evolution of multicellular organisms, cells are associated with forming tissues. For example, most animals have muscle tissue and nervous tissue.

- In most complex organisms, tissues organise into functional structures called organs, such as the heart and stomach in animals and roots and leaves in plants.



Hierarchy in organisation of an organism (Vertical Scale)

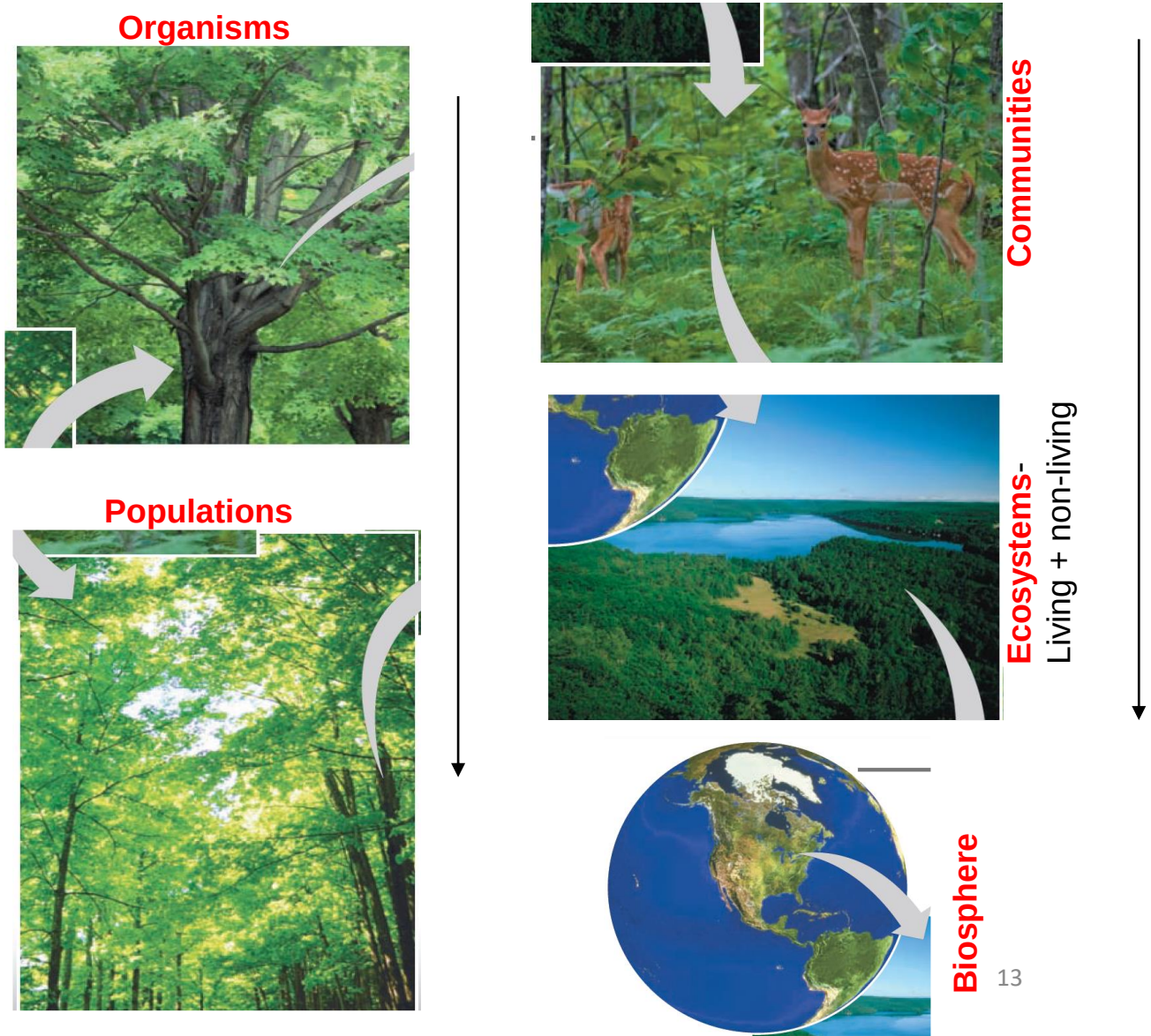
- Functioning together with great precision, organ systems comprise a complex, multicellular organism. Organisms interact to form still more complex levels of biological organisation. All the members of one species living in the same geographic area simultaneously comprise a population.

- The populations of various organisms that inhabit a particular area and interact with one another form a community.

- A community, together with its nonliving environment, is an ecosystem. An ecosystem can be as small as a pond (or even a puddle) or as vast as the Great Plains of North America or the Arctic tundra.

- All of Earth's ecosystems together are known as the biosphere. The biosphere includes all of Earth inhabited by living organisms—the atmosphere, the hydrosphere (water in any form), and the lithosphere (Earth's crust).

- The study of how organisms relate to one another and to their physical environment is called ecology.



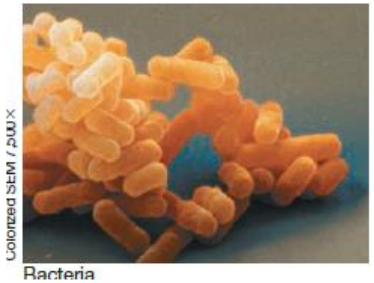
Three domains of life

Domains of life

Simple cells

Complex cells

Prokarya



Bacteria

Archea

Archaea live in Earth's extreme environments, such as salty lakes and boiling hot springs.

Eukarya

Single cells

Distinguished partly by their modes of nutrition

Protists

Plante

Which produce their own food by photosynthesis.

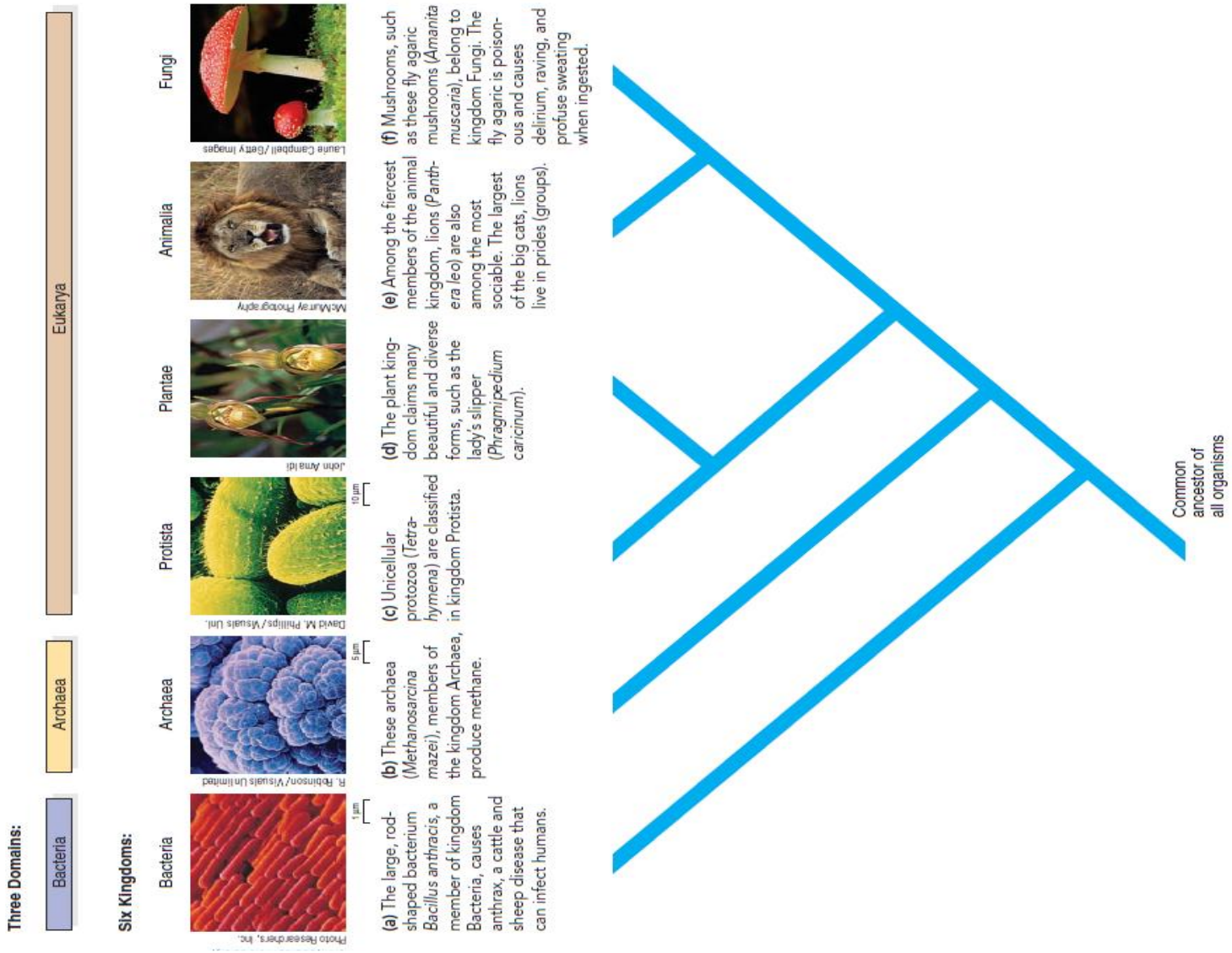
Fungi

Mostly decompose organic wastes and absorb the nutrients into their cells

Animalia

Animalia, obtain food by eating other organisms

Tree of Life: Molecular evidence



- These three multicellular kingdoms are distinguished partly by how the organisms obtain food.

1. Plants produce their own sugars and other foods by photosynthesis.
2. Fungi are mostly decomposers, digesting dead organisms and organic wastes.
3. Animals obtain food by ingesting (eating) and digesting other organisms.

- Those eukaryotes that do not fit into any of the three kingdoms fall into a catch-all group called the protists.

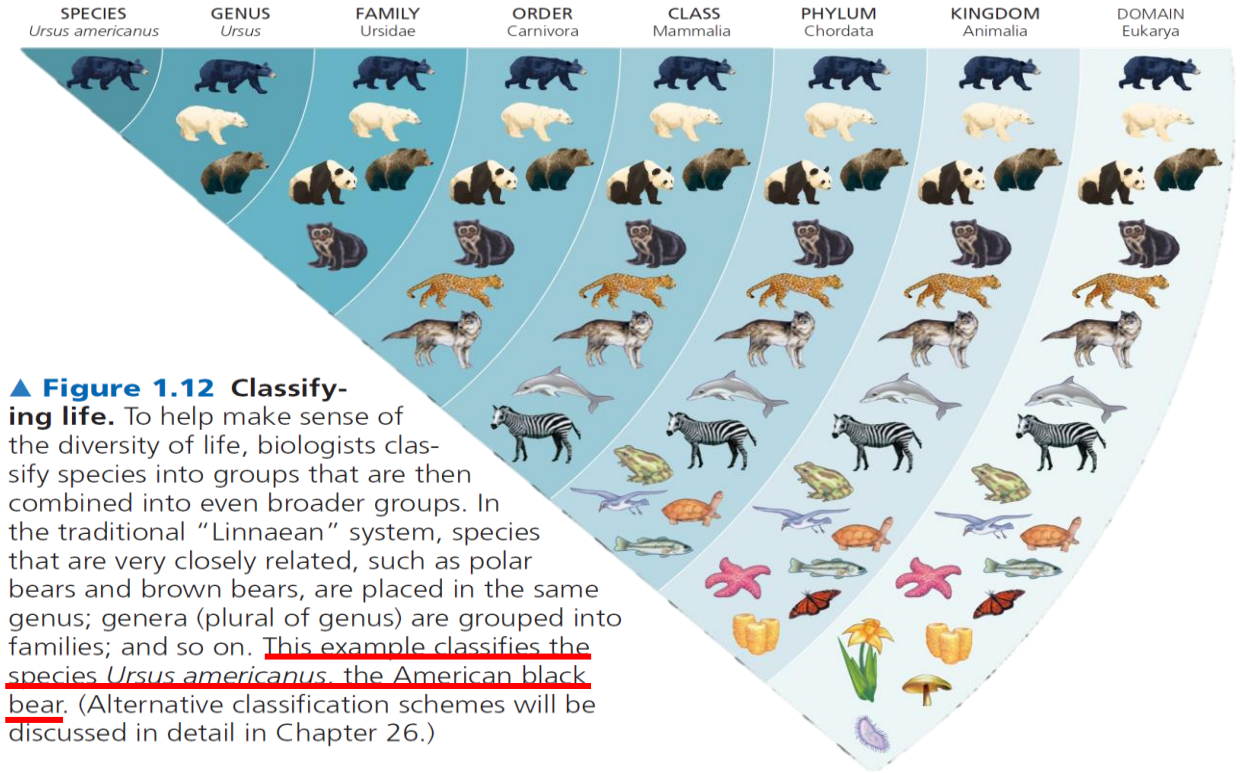
1. Most protists are single-celled; they include microscopic organisms such as amoebas.
2. But protists also include certain multicellular forms, such as seaweeds.
3. Scientists are in the process of organizing protists into multiple kingdoms, although they do not yet agree on exactly how to do this.

How do biologists name and arrange organisms? (Horizontal Scale: Life's past, present and future)

To study life, we need a system for organizing, naming, and classifying its myriad forms.

Systematics is the field of biology that studies the diversity of organisms and their evolutionary relationships.

Taxonomy, a subspecialty of systematics, is the science of naming and classifying organisms.



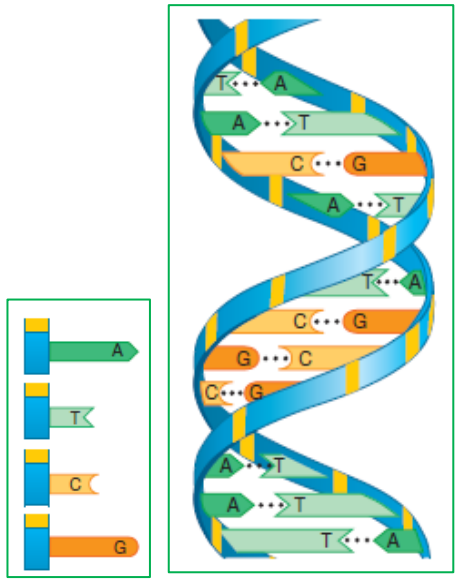
▲ **Figure 1.12 Classifying life.** To help make sense of the diversity of life, biologists classify species into groups that are then combined into even broader groups. In the traditional "Linnaean" system, species that are very closely related, such as polar bears and brown bears, are placed in the same genus; genera (plural of genus) are grouped into families; and so on. This example classifies the species *Ursus americanus*, the American black bear. (Alternative classification schemes will be discussed in detail in Chapter 26.)

Category	Cat	Human	White Oak
Domain	Eukarya	Eukarya	Eukarya
Kingdom	Animalia	Animalia	Plantae
Phylum	Chordata	Chordata	Anthophyta
Subphylum	Vertebrata	Vertebrata	None
Class	Mammalia	Mammalia	Eudicotyledones
Order	Carnivora	Primates	Fagales
Family	Felidae	Hominidae	Fagaceae
Genus	*Felis*	*Homo*	*Quercus*
Species	*Felis catus*	*Homo sapiens*	*Quercus alba*

Biologists use a hierarchical classification scheme with a series of taxonomic categories from species to domain; each category is more general and more inclusive than the one below it.

Theme-1: Life depends on the flow of information

- For an organism to grow, develop, carry on self-regulated metabolism, respond to stimuli, and reproduce, it must have precise instructions and its cells must be able to communicate.
- The information an organism requires to carry on these life processes is coded and delivered in the form of chemical substances and electrical impulses.
- Genetic code in DNA determines an organism's structures and functions.
- Where, when, and how an organism's genetic information is expressed is also dependent on external stimuli.
- External stimuli such as light, sound, or chemicals, and internal stimuli such as food in your stomach or an excess of sugar in your blood.
- This external stimuli affects gene expression.
- Indeed, the integrated flow of genetic and other types of information is essential for life.



Nucleotide to DNA structure

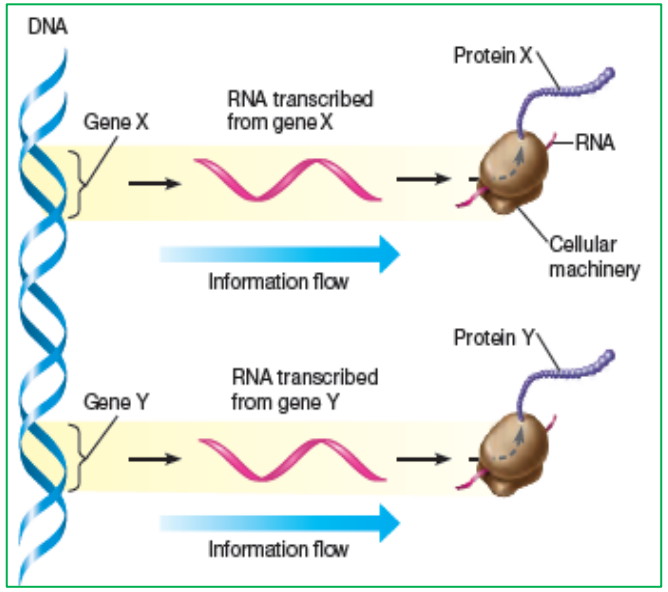
Emergent property at molecular level

Flow of information is unidirectional

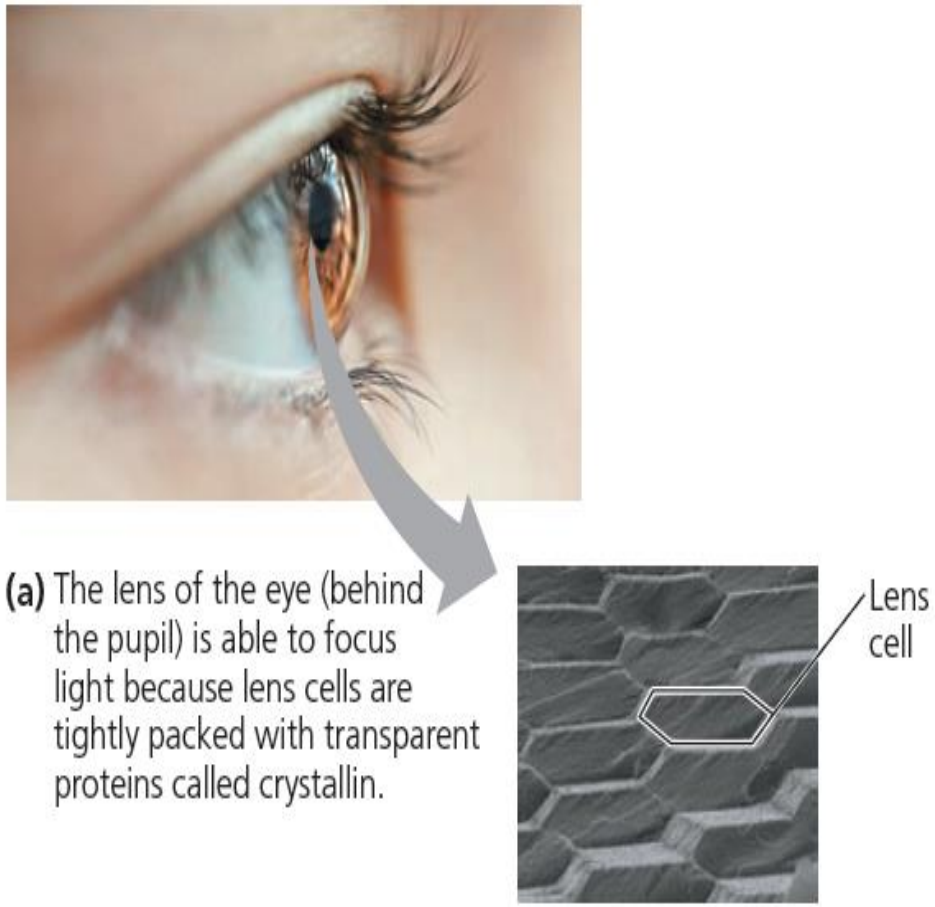
i.e., DNA
transcription ↓
RNA
translation ↓
Protein

Genetic code

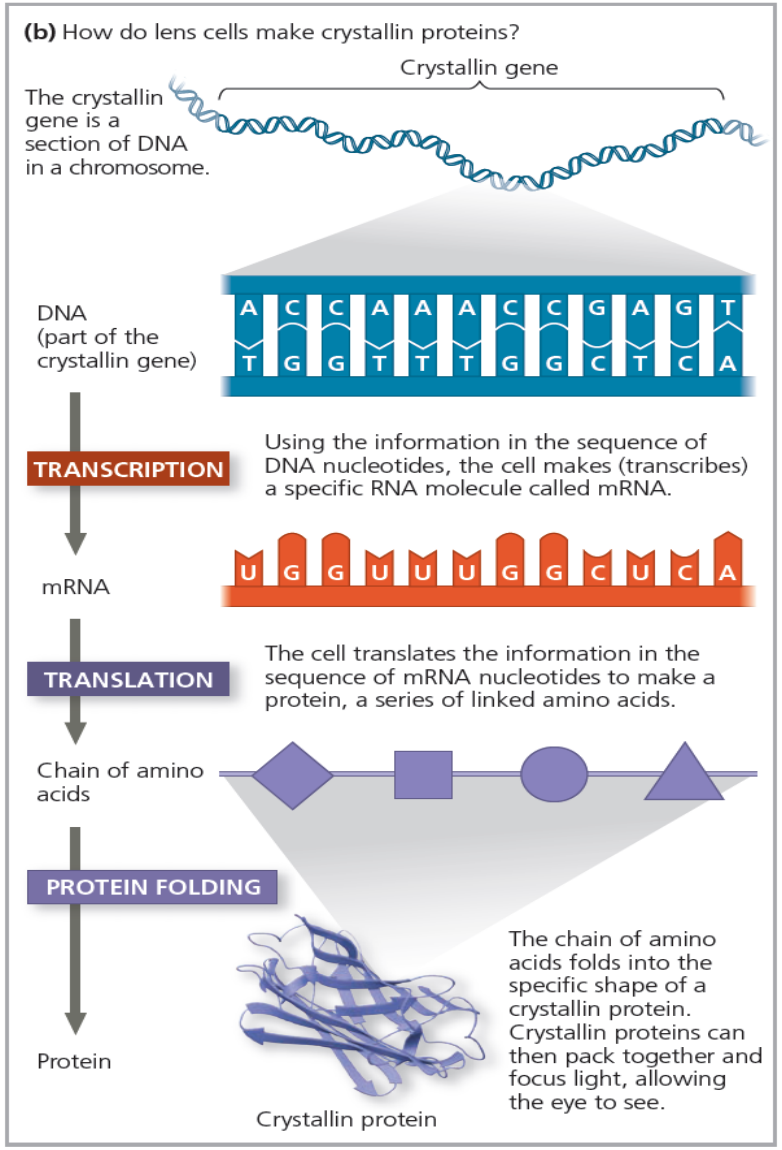
Gene Expression



Theme-2: Structure and function: Producing lens protein in the eyes.



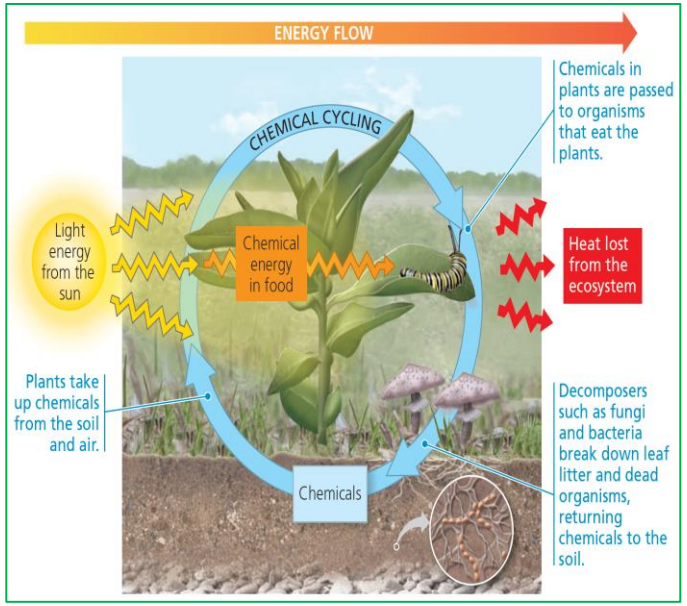
(a) The lens of the eye (behind the pupil) is able to focus light because lens cells are tightly packed with transparent proteins called crystallin.

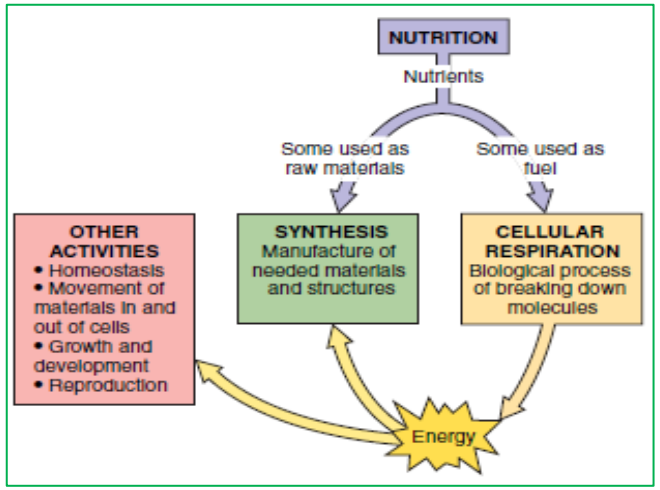


At the molecular level, the structure of a protein correlates with its function

Theme-3: Energy is another important theme of life

- Life depends on continuous energy input from the sun because every activity of a living cell or organism requires energy.
- Whenever energy is used to perform biological work, some is converted to heat and dispersed into the environment.
- All the energy transformations and chemical processes that occur within an organism are referred to as its *metabolism*.
- Certain nutrients are used as fuel for *cellular respiration*, a process that releases some of the energy stored in the nutrient molecules and use this to do work.
- Like individual organisms, ecosystems depend on a continuous input of energy.





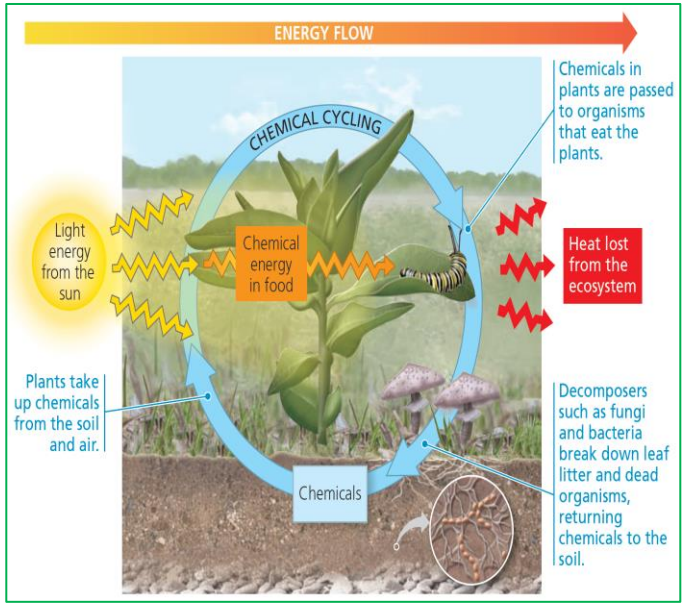
Energy is another important theme of life

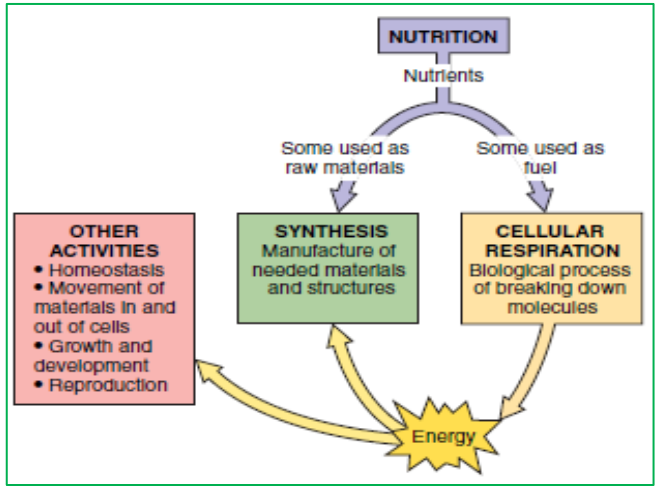
- A self-sufficient ecosystem contains *three types of organisms*—producers, consumers, and decomposers— and a physical environment where they can survive.

- Producers, or autotrophs, are plants, algae, and certain bacteria that produce food from simple raw materials.

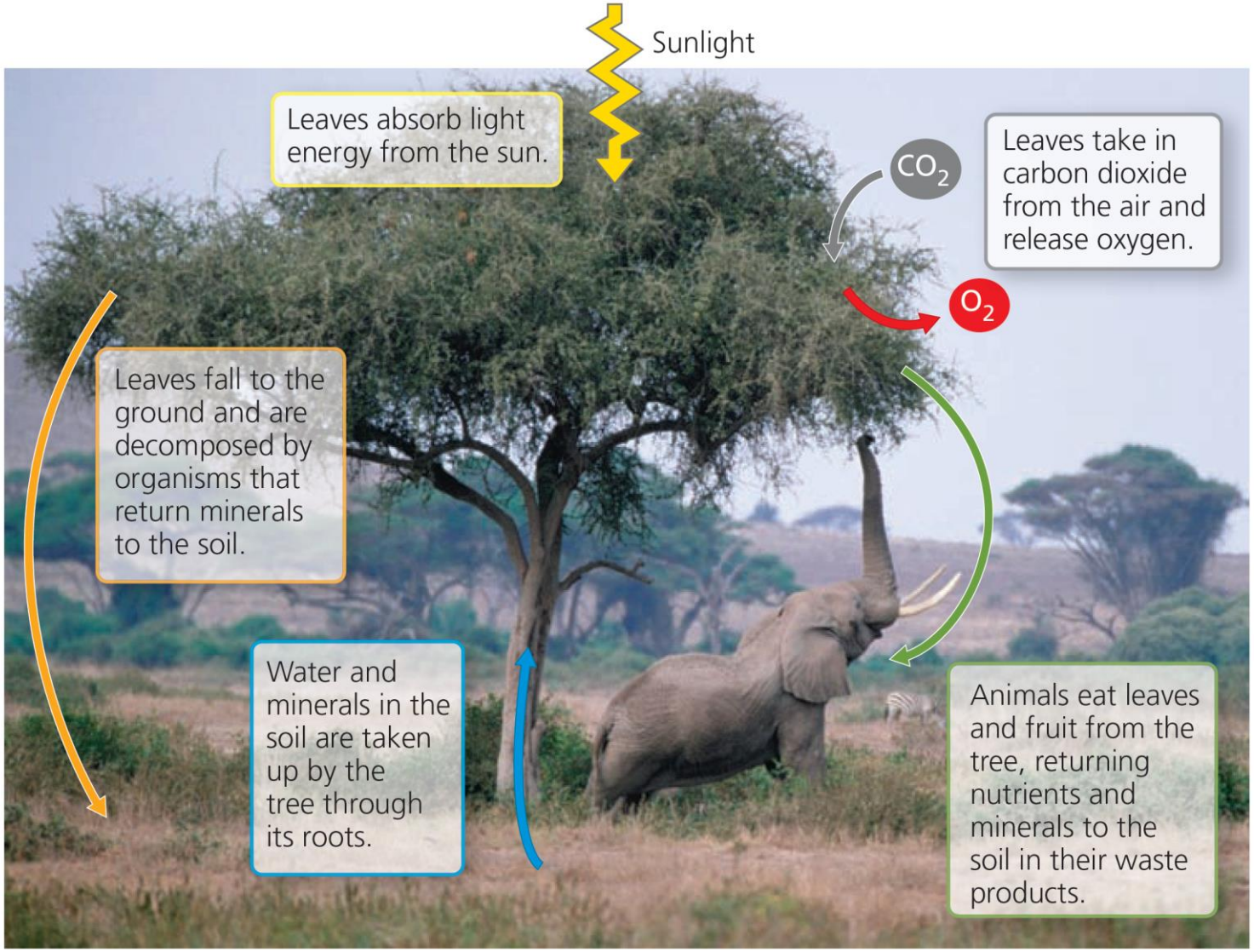
- Animals are consumers, or heterotrophs—organisms that depend on producers for food, energy, and oxygen.

- Most bacteria and fungi are decomposers, heterotrophs that obtain nutrients by breaking down nonliving organic material such as wastes, dead leaves and branches, and the bodies of dead organisms.





Hierarchy of interactions in an ecosystem



Interaction

Theme-4: Molecular interactions

❖ Illustration of external signal regulating gene expression: Case of insulin production.

❖ All body cells have the same genetic information, but the gene for insulin is only expressed in certain cells in your pancreas. It broadly involves 4 steps.

1. After a meal, the level of the sugar glucose in your blood rises

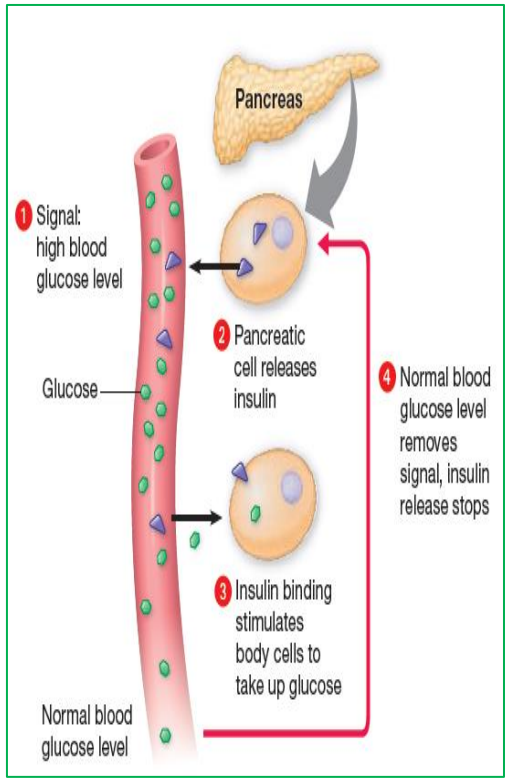
2. This internal signal stimulates cells in your pancreas to secrete the hormone insulin, which travels throughout your body in your blood.

3. Insulin binds to receptors on body cells, causing them to take up glucose.

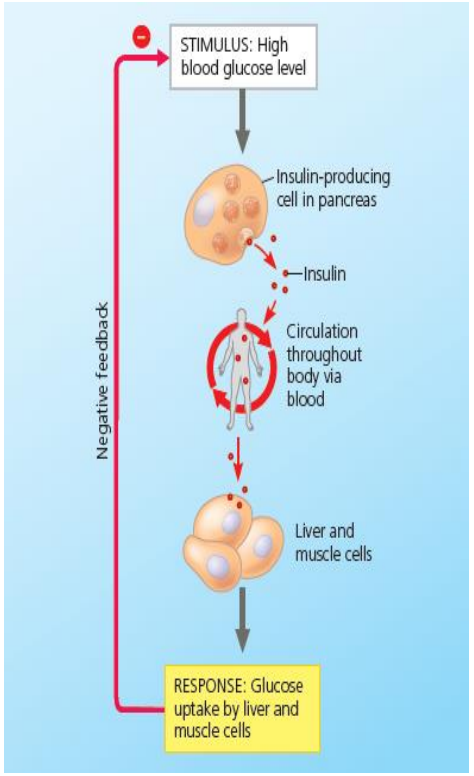
4. The now-lowered blood glucose level removes the signal, and insulin secretion decreases.

❖ Information flow through such regulatory systems enables organisms to maintain relatively stable and beneficial internal conditions.

Regulation of glucose by insulin



Negative feedback mechanism



Theme-5: Evolution is the core theme of biology

- **Evolution** is the process of change that has transformed life on Earth from its earliest forms to the vast array of organisms living today.

- Influential work on evolution was published by Charles Darwin in 1859 on his book entitled *On the Origin of Species by Means of Natural Selection.*

Two key points presented in the book are

1. Descent with modification.

2. Natural selection

Example for point-1

The flamingo, penguin, and hummingbird all have a common "bird" body plan of wings, beak, feet, and feathers, but these structures are highly specialized for each bird's unique lifestyle.

Darwin's theory of natural selection

- Every organism is the product of numerous interactions between environmental conditions and the genes inherited from its ancestors. Organism = Environment x Genes

- If all individuals of a species were exactly alike, any change in the environment might be disastrous to all, and the species would become extinct. (importance of organism variations within populations)

- Adaptations to changes in the environment occur as a result of evolutionary processes that take place over time and involve many generations. (Evolution through adaptation to environment)

- Charles Darwin and Alfred Wallace were the first to brought a theory of evolution to general attention and suggested a plausible mechanism, natural selection.

- They put forward two points:

  1. *Species living today arose from a succession of ancestors that were different from them. Darwin called this process "descent with modification."*

- The above insightful phrase captures both the unity of life (descent from a common ancestor) and the diversity of life (modifications that evolved as species diverged from their ancestors).

Natural selection: second observation

2. Darwin's mechanism of evolution is through **natural selection**.
   Darwin started with two observations, from which he drew two inferences.

Observation #1: *Individual variation*. Individuals in a population vary in their traits, many of which seem to be heritable (passed on from parents to offspring).

Observation #2: *Overproduction of offspring*. All species can produce far more offspring than the environment can support. Competition for resources is thus inevitable, and many of these offspring fail to survive and reproduce.

Inference #1: *Unequal reproductive success*. Individuals with inherited traits best suited to the local environment are more likely to survive and reproduce than are less well suited individuals.

Inference #2: *Accumulation of favorable traits overtime*. As a result of this unequal reproductive success over many generations, a higher and higher proportion of individuals in the population will have the advantageous traits.

1 Population with varied inherited traits.

2 Elimination of individuals with certain traits and reproduction of survivors.

3 Increasing frequency of traits that enhance survival and reproductive success.

Summary of Darwin's theory

- To summarize, Darwin based his theory of natural selection on the following four observations:

1. *Individual members of a species show some variation from one another.*

2. *Organisms produce many more offspring than will survive to reproduce.*

3. *Organisms compete for necessary resources such as food, sunlight, and space. Individuals with characteristics that enable them to obtain and use resources are more likely to survive to reproductive maturity and produce offspring.*

4. *The survivors that reproduce pass their adaptations for survival on to their offspring. Thus, the best-adapted individuals of a population leave, on average, more offspring than do other individuals.*

- Scientists now understand that most variations among individuals are a result of different varieties of genes that code for each characteristic.

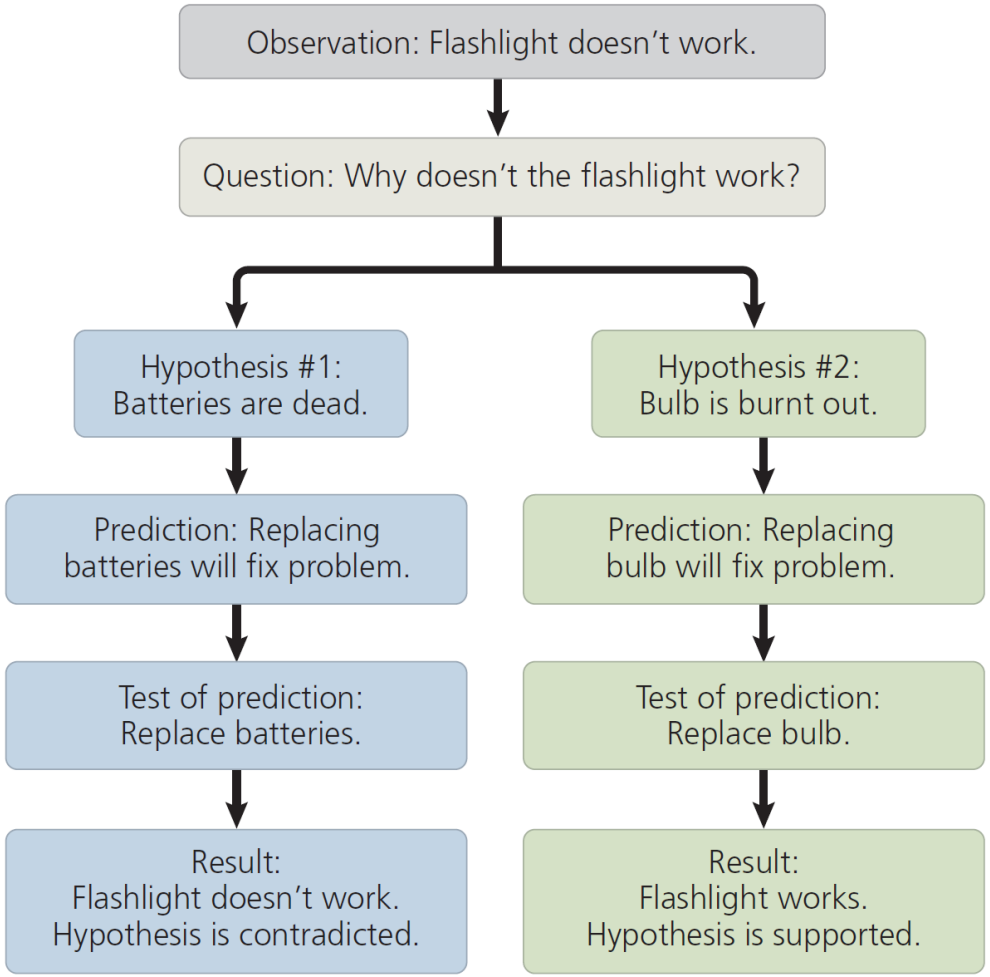
- The ultimate source of these variations is random mutations, chemical or physical changes in DNA that persist and can be inherited. Mutations modify genes and by this process provide the raw material for evolution

Tutorials: Hypotheses, theories, data analysis

Role of science and technology

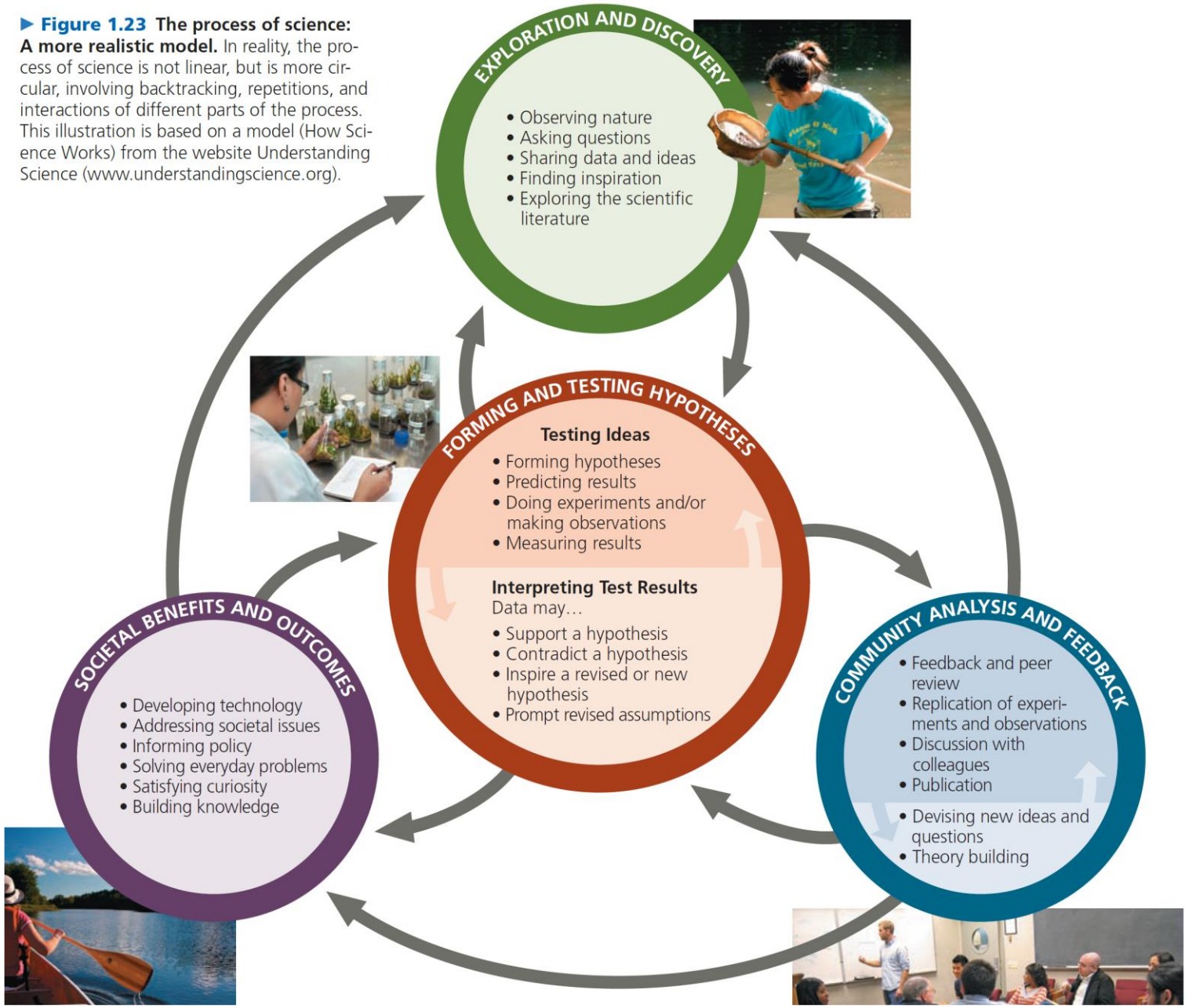
Scientific inquiry

- **Science** is a way of knowing—an approach to understanding the natural world.
- **Hypothesis**: Logical, testable explanations
- Deductive Reasoning



▲ **Figure 1.22 A simplified view of the scientific process.** The idealized process sometimes called the "scientific method" is shown in this flow chart, using a campground example of hypothesis testing.

▶ Figure 1.23 The process of science: A more realistic model. In reality, the process of science is not linear, but is more circular, involving backtracking, repetitions, and interactions of different parts of the process. This illustration is based on a model (How Science Works) from the website Understanding Science (www.understandingscience.org).



Explanation

1. data 2. qualitative 3. inductive 4. observation 5. question 6. hypothesis 7. testable 8. falsifiable 9. prove 10. theory 11. experiment 12. quantitative 13. experimental 14. control 15. factor 16. prediction 17. deductive

- At the heart of science is inquiry, a search for information and explanations of natural phenomena.
- Scientists use a process of inquiry that includes making observations, forming logical, testable explanations (hypotheses), and testing them.
- Recorded observations are called data. Put another way, data are items of information on which scientific inquiry is based.
- The term data implies numbers to many people. But some data are qualitative, often in the form of recorded descriptions rather than numerical measurements
- Collecting and analyzing observations can lead to important conclusions based on a type of logic called inductive reasoning.
- In science, a hypothesis is a tentative answer to a well framed question—an explanation on trial.
- An experiment is a scientific test, carried out under controlled conditions.

Extra slides

To add more

▼ Figure 1.3

Exploring Levels of Biological Organization

◀ 1 The Biosphere

Even from space, we can see signs of Earth's life—in the green mosaic of the forests, for example. We can also see the scale of the entire biosphere, which consists of all life on Earth and all the places where life exists: most regions of land, most bodies of water, the atmosphere to an altitude of several kilometers, and even sediments far below the ocean floor.

◀ 2 Ecosystems

Our first scale change brings us to a North American forest with many deciduous trees (trees that lose their leaves and grow new ones each year). A deciduous forest is an example of an ecosystem, as are grasslands, deserts, and coral reefs. An ecosystem consists of all the living things in a particular area, along with all the nonliving components of the environment with which life interacts, such as soil, water, atmospheric gases, and light.

▶ 3 Communities

The array of organisms inhabiting a particular ecosystem is called a biological community. The community in our forest ecosystem includes many kinds of trees and other plants, various animals, mushrooms and other fungi, and enormous numbers of diverse microorganisms, which are living forms, such as bacteria, that are too small to see without a microscope. Each of these forms of life is called a *species*.

▶ 4 Populations

A population consists of all the individuals of a species living within the bounds of a specified area. For example, our forest includes a population of sugar maple trees and a population of white-tailed deer. A community is therefore the set of populations that inhabit a particular area.

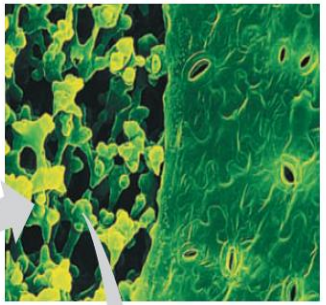
▲ 5 Organisms

Individual living things are called organisms. Each of the maple trees and other plants in the forest is an organism, and so is each deer, frog, beetle, and other forest animals. The soil teems with microorganisms such as bacteria.

1-6 can viewed with our eyes.

▼ 6 Organs and Organ Systems

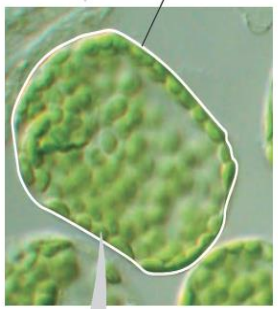
The structural hierarchy of life continues to unfold as we explore the architecture of more complex organisms. A maple leaf is an example of an organ, a body part that carries out a particular function in the body. Stems and roots are the other major organs of plants. The organs of complex animals and plants are organized into organ systems, each a team of organs that cooperate in a larger function. Organs consist of multiple tissues.



◀ 7 Tissues

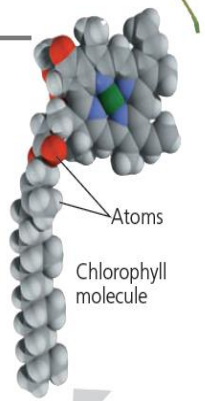
Viewing the tissues of a leaf requires a microscope. Each tissue is a group of cells that work together, performing a specialized function. The leaf shown here has been cut on an angle. The honeycombed tissue in the interior of the leaf (left side of photo) is the main location of photosynthesis, the process that converts light energy to the chemical energy of sugar. The jigsaw puzzle–like "skin" on the surface of the leaf is a tissue called epidermis (right side of photo). The pores through the epidermis allow entry of the gas CO_2, a raw material for sugar production.

7-10 requires microscopes or similar instruments to view.



▶ 10 Molecules

Our last scale change drops us into a chloroplast for a view of life at the molecular level. A molecule is a chemical structure consisting of two or more units called atoms, represented as balls in this computer graphic of a chlorophyll molecule. Chlorophyll is the pigment molecule that makes a maple leaf green, and it absorbs sunlight during photosynthesis. Within each chloroplast, millions of chlorophyll molecules are organized into systems that convert light energy to the chemical energy of food.



▲ 8 Cells

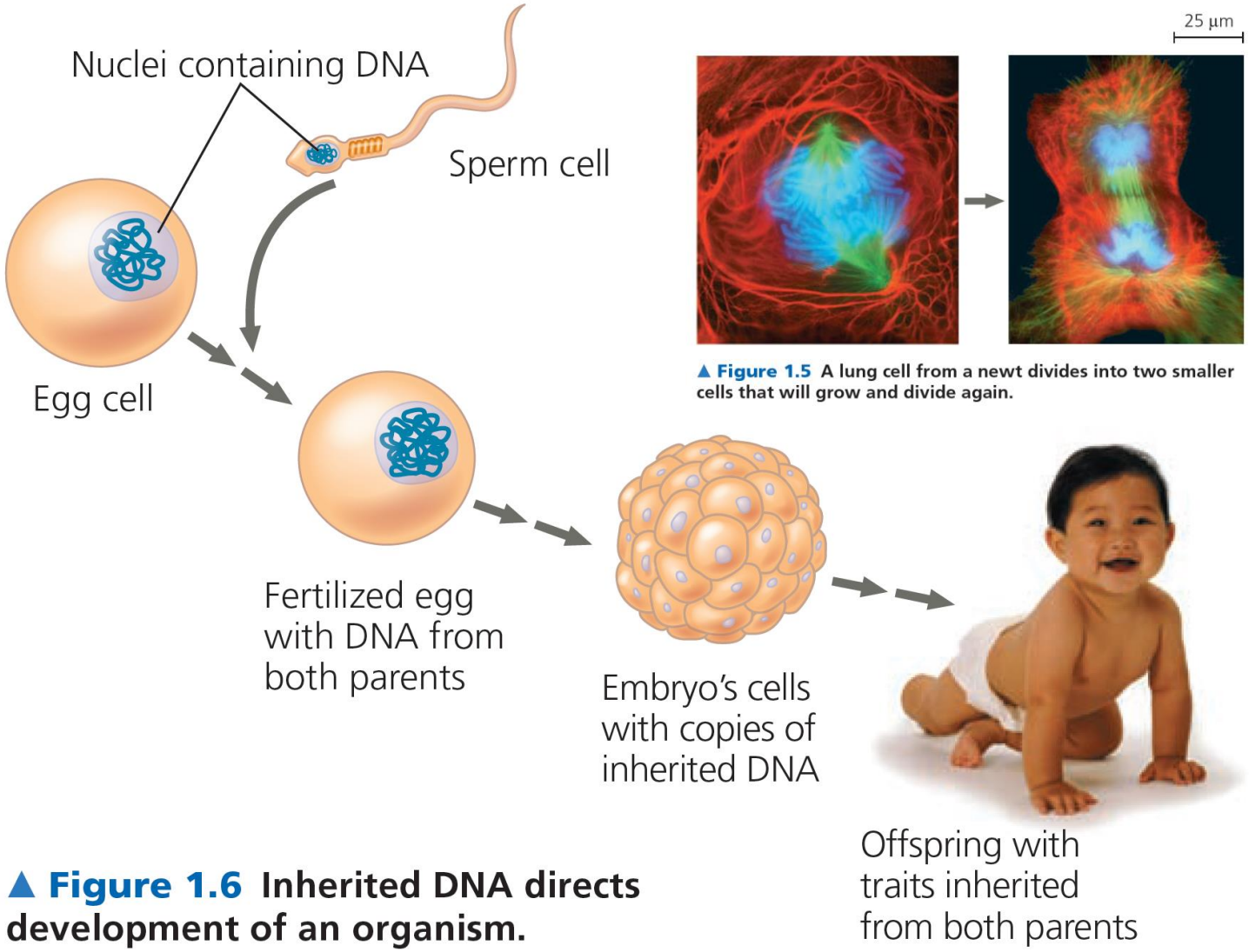
The cell is life's fundamental unit of structure and function. Some organisms are single cells, while others are multicellular. A single cell performs all the functions of life, while a multicellular organism has a division of labor among specialized cells. Here we see a magnified view of cells in a leaf tissue. One cell is about 40 micrometers (µm) across—about 500 of them would reach across a small coin. As tiny as these cells are, you can see that each contains numerous green structures called chloroplasts, which are responsible for photosynthesis.

▶ 9 Organelles

Chloroplasts are examples of organelles, the various functional components present in cells. This image, taken by a powerful microscope, shows a single chloroplast.

Organization

Expression and transmission of genetic information



▲ **Figure 1.5** A lung cell from a newt divides into two smaller cells that will grow and divide again.

▲ **Figure 1.6 Inherited DNA directs development of an organism.**

Continued..

- Cells vary greatly in size and appearance, all organisms consist of basic units called cells.
- New cells are formed only by the division of previously existing cells. These concepts are expressed in the cell theory.
- Some of the simplest life-forms, such as protozoa, are unicellular organisms, meaning that each consists of a single cell.
- In contrast, the body of a cat or a maple tree is made of billions of cells. In such complex multicellular organisms.
- Life processes depend on the coordinated functions of component cells that may be organized to form tissues, organs, and organ systems.
- Every cell is enveloped by a protective plasma membrane that separates it from the surrounding external environment. The plasma membrane regulates passage of materials between cell and environment.
- Cells have specialized molecules that contain genetic instructions and transmit genetic information. In most cells, the genetic instructions are encoded in deoxyribonucleic acid, more simply known as DNA.
- Cells typically have internal structures called organelles that are specialized to perform specific functions.